NUTM
NIGERIAN UNIVERSITY OF TECHNOLOGY & MANAGEMENT

CYB 305

Digital Forensics & Investigation Methods

Course Overview

Dr. Segun Popoola

Senior Lecturer in Cybersecurity and AI

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Dr. Segun Popoola

Lecturer for CYB 305.



Senior Lecturer in Cyber Security & Artificial Intelligence

Anglia Ruskin University, Cambridge, United Kingdom

2015 | BTech

Electronic and Electrical Engineering (*First Class*)

Ladoke Akintola University of Technology, Ogbomoso, Nigeria

2018 | MEng

Information and Communication Engineering (*Distinction*)

Covenant University, Ota, Nigeria

2022 | PhD

Cybersecurity and Artificial Intelligence

Manchester Metropolitan University, United Kingdom

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Course questions and contact details

Use the official NUTM / Teams channels for course questions.

During class

Use Raise Hand, the meeting chat, or a brief verbal question. You can respond without switching on your camera.

Between classes

Check the official NUTM channel for announcements and contact arrangements.

Email: segunpopoola15@gmail.com

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Credits and contact hours

The course includes lectures and practical investigation work.

2

credit units

15

lecture hours

45

practical hours

Total contact time: 60 hours.

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Sources of digital evidence

Use the Teams chat to add your examples.

Where might you find digital evidence?

01

A phone

02

A bank-app record

03

A website log

MICROSOFT TEAMS CHAT

Type another source in the chat.

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Digital evidence in Nigerian settings

Fictional scenarios for class discussion.

1 DIGITAL PAYMENTS

A customer disputes a transfer after receiving a suspicious message.

2 MESSAGING & SOCIAL

A WhatsApp account is impersonated to solicit money from contacts.

3 ORGANISATIONS

Confidential company data appears outside its authorised location.

4 PUBLIC SERVICES

A school or government portal shows signs of unauthorised access.

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Course learning outcomes

The course has seven learning outcomes.

01 Explain core digital-forensics and evidence concepts

02 Recognise challenges in collecting and interpreting evidence

03 Appraise cyber trails and reconstruct activity carefully

04 Use the terminology of computer-crime investigations

05 Connect technology, crime, law, privacy and jurisdiction

06 Examine the techniques and tools investigators use

07 Follow best practice across the evidence lifecycle

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Semester topic sequence

A 12-week topic sequence; exact delivery dates will follow the NUTM timetable.

WEEKS 1–3

FOUNDATIONS & ACQUISITION

Evidence, chain of custody, storage and imaging

WEEKS 4–6

ENDPOINTS & RECONSTRUCTION

Windows artefacts, deleted data, timelines

WEEKS 7–9

SPECIALIST SOURCES

Memory, network, email and mobile artefacts

WEEKS 10–12

LAW, CASEWORK & REPORTING

Nigeria-centred legal context; integrated investigation

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Forensic tools for practical sessions

Availability depends on approved lab resources and tool licensing.

1 Autopsy / Sleuth Kit

Disk and file-system examination

2 FTK Imager

Acquisition and verification

3 Wireshark

Network traffic analysis

4 Volatility 3

Memory artefact examination

5 ExifTool

Metadata inspection

6 Hash utilities

Integrity checks and verification

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Class activities on Microsoft Teams

Sessions include explanations, case discussions and practical demonstrations.

1	2	3	4	5
EXPLANATION	**CASE STUDY**	**QUESTIONS**	**LAB WORK**	**FINDINGS**
Brief concept explanations	Pause to interpret a case	Poll, chat or Raise Hand	Guided practical demonstration	Report a finding and explain its limitations.

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What does a screenshot establish?

Respond A, B or C in the Teams chat.

A screenshot shows that a bank transfer happened. What can we conclude?

A It proves who authorised the transfer.

B It is a lead that needs independent verification.

C It should be edited to remove inconvenient details.

Check where the file came from, whether it changed and what it records.

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Roles in digital forensics

Some roles require additional training or professional experience.

Digital Forensics Analyst

Examine and report digital artefacts

Cybercrime Investigator

Support investigations with evidential methods

Incident Response Analyst

Reconstruct breaches and triage systems

Mobile Forensics Examiner

Analyse smartphone evidence lawfully

Malware Analyst

Understand suspicious software behaviour

eDiscovery / Forensic Consultant

Manage electronic evidence for organisations

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Employers and professional settings

Examples of organisations and sectors, not current vacancies.

PUBLIC INVESTIGATIONS

Law-enforcement and financial-crime investigation teams (e.g., EFCC, Nigeria Police).

FINANCE & FINTECH

Banks, payment platforms and fraud-response / digital-risk teams.

TELECOMS & TECHNOLOGY

Operators, service providers, cloud and enterprise security teams.

CONSULTANCY & LEGAL

Cybersecurity consultancies, forensic laboratories and eDiscovery support.

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Preparing for forensic work

Keep records of your lab work that another examiner can follow.

01	**Study evidence handling**	Practise integrity checks and document each examination step.
02	**Keep sample case reports**	Include lab notes, hashes and the basis for your conclusions.
03	**Choose an area to study**	Compare disk, memory, mobile and network examination.
04	**Look for supervised work**	Consider internships, university labs and graduate roles.

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Evidence handling and reporting

Use authorised training data and follow applicable Nigerian law.

PRESERVE

Work from forensic copies; document decisions.

PROTECT

Respect privacy and confidentiality.

VERIFY

Check hashes, artefacts and sources; do not fabricate findings.

REPORT

Separate observations from interpretations.

Nigeria: Evidence Act 2011 • Cybercrimes Act 2015 (as amended in 2024) • Nigeria Data Protection Act 2023

Case discussion: a disputed transfer

Fictional scenario: a Lagos fintech customer contests an account transfer.

Available: a phone, a transaction screenshot and a server-log excerpt.

1. What other evidence might exist?
2. What must you preserve first?
3. What cannot the screenshot prove by itself?